Formula 1

Telemetry Insights Visual Dashboard



Project Report By:

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Problem

Formula 1 is one of the most data-driven sports, yet driver impact is often overlooked

- Engineers want to identify optimal lap patterns
- Drivers benefit from objective performance feedback

What is hard about F1 telemetry analysis?

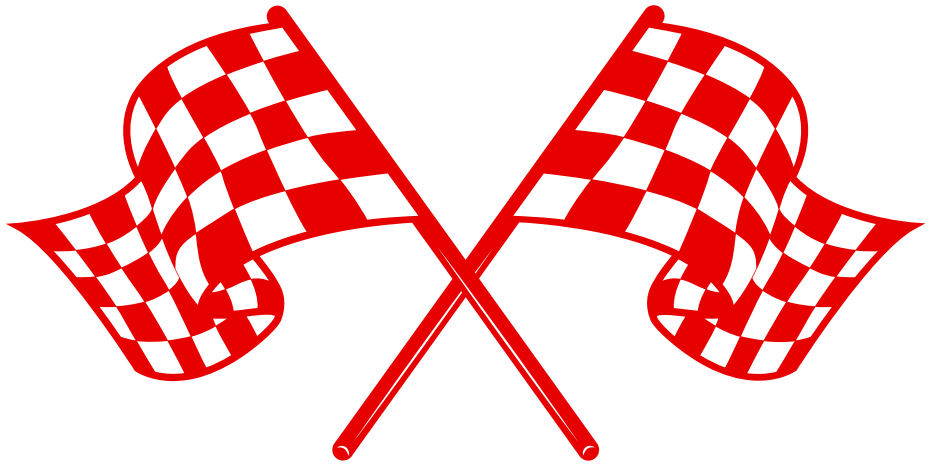
- Raw lap data is massive, multidimensional, and difficult to interpret
- Hard to identify patterns in driver performance manually
- Need automated way to group similar laps and identify driving styles



Objective

- Transform raw F1 telemetry into meaningful driver insights
- Identify performance patterns through clustering algorithms
  - Group drivers into "driving styles"
- Develop an interactive dashboard for intuitive data exploration
- Given the choice between two open source F1 data sources
  - Tracing Insights vs FastF1





Data Ingestion

Initial Tracing Insights Data

Telemetry output for a single lap


{"tel":{"time":
[0.0,0.041,0.06,0.22,0.321,0.481,0.52,0.681,0.82,1.02,1.081,1.361,1.4,1.6,1.601,1.9,1.921,2.16,2.241,2.38,2.521,2.54,2.72,2.881,2.92,3.241,3.26,3.44,3.481,3.64,3.92,4.06,4.16,4.18,4.44,4.5,4.68,4.981,5.0,5.2,5.281,5.44,5.6,5.621,5.781,5.88,6.101,6.121,6.34,6.5,6.561,6.86,6.881,7.0,7.041,7.38,7.401,7.601,7.64,7.761,7.86,8.121,8.18,8.281,8.6,8.601,8.881,9.02,9.08,9.121,9.401,9.42,9.601,9.76,9.801,9.98,10.081,10.241,10.26,10.4,10.52,10.56,10.78,10.84,11.08,11.16,11.3,11.52,11.561,11.72,11.761,11.92,12.061,12.24,12.361,12.481,12.561,12.681,12.84,12.921,13.1,13.321,13.34,13.56,13.601,13.76,13.761,13.921,14.0,14.161,14.18,14.321,14.44,14.521,14.68,14.721,14.98,15.001,15.16,15.161,15.36,15.361,15.54,15.561,15.86,15.881,16.16,16.281,16.48,16.68,16.681,16.92,17.001,17.22,17.24,17.52,17.72,17.9,17.92,18.141,18.241,18.32,18.481,18.56,18.661,18.88,18.901,19.121,19.301,19.361,19.46,19.601,19.76,19.881,19.96,20.18,20.201,20.361,20.4,20.641,20.72,20.881,21.02,21.161,21.32,21.401,21.62,21.8,21.841,21.96,22.041,22.18,22.201,22.441,22.54,22.7,22.801,23.041,23.08,23.241,23.28,23.46,23.561,23.8,23.9,24.08,24.12,24.32,24.661,24.68,24.841,24.92,25.041,25.12,25.161,25.4,25.421,25.641,25.761,25.901,26.041,26.26,26.281,26.441,26.48,26.641,26.66,26.82,26.881,27.121,27.18,27.34,27.361,27.641,27.72,27.841,28.0,28.121,28.26,28.321,28.48,28.601,28.801,28.84,29.001,29.28,29.401,29.54,29.721,29.82,29.98,30.121,30.16,30.281,30.44,30.52,30.62,30.78,30.84,31.02,31.04,31.261,31.32,31.48,31.521,31.68,31.701,31.921,31.96,32.141,32.241,32.281,32.481,32.58,32.721,32.78,32.96,32.961,33.24,33.241,33.441,33.48,33.721,33.84,33.961,34.121,34.2,34.3,34.481,34.6,34.761,34.86,35.14,35.201,35.34,35.401,35.5,35.641,35.84,35.841,36.121,36.16,36.3,36.361,36.52,36.521,36.761,36.88,37.081,37.2,37.44,37.48,37.741,37.88,37.921,38.08,38.261,38.32,38.361,38.48,38.661,38.821,38.881,39.021,39.081,40.04,40.121,40.32,40.361,40.48,40.561,40.66,40.841,41.0,41.121,41.321,41.36,41.58,41.721,41.9,42.001,42.22,42.32,42.441,42.56,42.8,42.801,43.02,43.281,43.34,43.56,43.681,43.82,43.841,44.12,44.14,44.32,44.361,44.48,44.701,44.76,44.941,45.101,45.12,45.481,45.561,45.921,45.94,46.1,46.281,46.36,46.441,46.48,46.641,46.72,47.0,47.041,47.28,47.281,47.56,47.561,47.721,47.74,47.92,47.961,48.22,48.321,48.64,48.641,48.801,48.84,49.001,49.08,49.201,49.28,49.32,49.521,49.68,49.86,49.881,50.04,50.121,50.36,50.521,50.68,50.72,50.88,51.04,51.161,51.28,51.321,51.48,51.501,51.72,51.92,51.921,52.081,52.141,52.281,52.62,52.641,52.881,52.9,53.14,53.161,53.321,53.4,53.56,53.681,53.78,53.96,53.961,54.16,54.161,55.0,55.001,55.2,55.361,55.5,55.521,55.74,55.801,55.9,56.08,56.081,56.281,56.4,56.44,56.521,56.8,56.841,57.02,57.121,57.4,57.42,57.621,57.72,57.941,58.04,58.181,58.24,58.4,58.421,58.641,58.761,58.9,58.961,59.161,59.24,59.361,59.44,59.64,59.681,59.881,59.9,60.041,60.1,60.241,60.5,60.521,60.681,60.74,60.92,60.961,61.3,61.361,61.52,61.561,61.801,62.02,62.161,62.28,62.441,62.48,62.7,62.721,63.04,63.081,63.281,63.36,63.56,63.681,63.88,64.081,64.241,64.36,64.461,64.541,64.6,64.741,64.76,65.0,65.041,65.161,65.201,65.421,65.441,65.56,65.601,65.921,65.94,66.081,66.12,66.32,66.321,66.54,66.64,66.681,66.96,67.001,67.18,67.241,67.401,67.56,67.62,67.681,67.921,67.98,68.161,68.24,68.56,68.561,68.66,68.801,69.001,69.04,69.241,69.42,69.561,69.6,69.68,69.721,69.881,69.9,70.1,70.201,70.28,70.441,71.101,71.16,71.301,71.32,71.501,71.52,71.801,71.961,72.1,72.121,72.3,72.401,72.561,72.64,72.78,72.841,73.0,73.161,73.22,73.641,73.66,73.841,73.94,74.001,74.14,74.42,74.481,74.681,74.7,74.921,75.02,75.2,75.321,75.52,75.561,75.68,75.94,75.961,76.201,76.28,76.481,76.5,76.681,76.74,76.881,76.94,77.201,77.241,77.441,77.68,77.701,78.0,78.121,78.361,78.501,78.601,78.66,78.761,78.9,79.14,79.161,79.3,79.361,79.48,79.74,79.761,80.04,80.241,80.44,80.521,80.64,80.68,80.841,81.08,81.081,81.361,81.38,81.54,81.601,81.82,81.881,81.98,82.081,82.18,82.321,82.48,82.58,82.601,82.761,82.9,83.081,83.1,83.241,83.441,83.5,83.641,83.66,83.961,84.4,84.401,84.68,84.801,84.921,84.978],"rpm":
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Database Design

Started with a

1 Table Architecture



Conglomerate Table

- Full Table Scan
- SELECT Statements
  - ~2 minutes per driver

Transitioned to a

2 Table Architecture

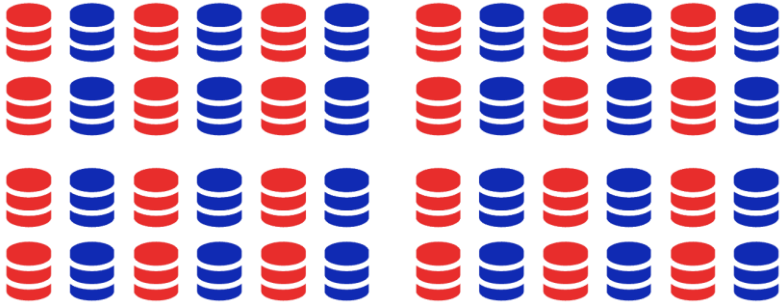


Metadata Table vs Telemetry Table

- Schema Rework
- SELECT Statements
  - ~1 minute per driver
  - 2x times speedup

Finalized a

48 Table Architecture



2 Tables (Metadata / Telemetry) Per Race

- Indexing / Table Rework
- SELECT Statements
  - ~1 second per driver
  - 60x speedup

Database Schema

```
postgres=# \d metadata_aus_2025
                Table "public.metadata_aus_2025"
   Column    |     Type     | Collation | Nullable | Default
-------------+--------------+-----------+----------+---------
 lap_data_id | integer      |           | not null |
 driver_id   | character(3) |           | not null |
 lap         | integer      |           | not null |
 tel_index   | integer      |           | not null |
Indexes:
    "metadata_aus_2025_pkey" PRIMARY KEY, btree (lap_data_id)
    "idx_metadata_aus_2025_driver_lap" btree (driver_id, lap)
Foreign-key constraints:
    "fk_telemetry" FOREIGN KEY (tel_index) REFERENCES telemetry_aus_2025(tel_index)
```

```
postgres=# \d telemetry_aus_2025
                          Table "public.telemetry_aus_2025"
      Column       |       Type       | Collation | Nullable |
-------------------+------------------+-----------+----------+---
 tel_index         | integer          |           | not null | ge
 time              | double precision |           | not null |
 distance          | double precision |           |          |
 rel_distance      | double precision |           |          |
 track_coordinate_x | double precision |           |          |
 track_coordinate_y | double precision |           |          |
 track_coordinate_z | double precision |           |          |
 rpm               | double precision |           |          |
 gear              | integer          |           |          |
 throttle          | double precision |           |          |
 brake             | boolean          |           |          |
 drs               | boolean          |           |          |
 speed             | double precision |           |          |
 acc_x             | double precision |           |          |
 acc_y             | double precision |           |          |
 acc_z             | double precision |           |          |
```

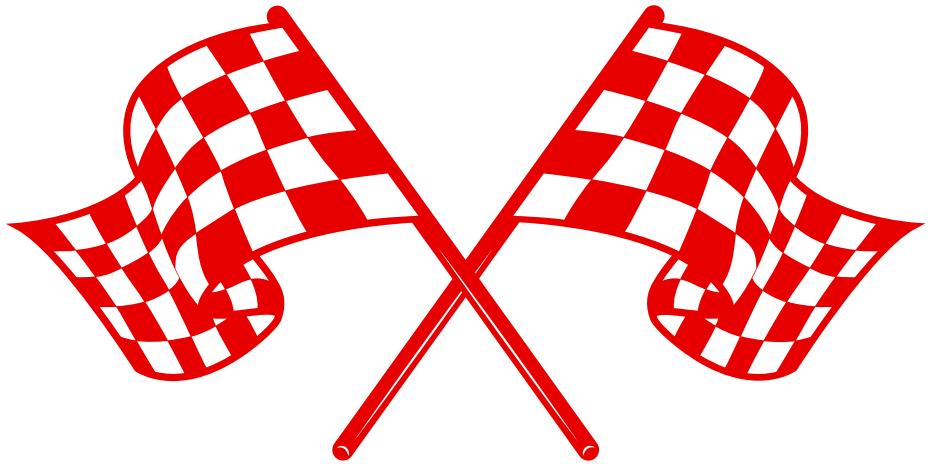
Indexing in Metadata for Fast Queries

~ **18,000,000** Total Telemetry Records!!

~ **750,000** Records Per Table

48 Total Tables





Data Processing

Metric Selection for Clustering

14 Total Telemetry Metrics

Throttle, RPM, Gear, Brake, Speed, Accel_x, Accel_y, Accel_z, Pos_x, Pos_y, Pos_z, Time, Rel_distance, and DRS

8 Selected Metrics

Throttle, RPM, Gear, Brake, Speed, Accel_x, Accel_y, Accel_z

Indexed by Relative Distance



Data Preprocessing



Data Reindexing

- Aligned all lap data to track distance
- Interpolated from existing data
- Grid of 500 points per lap

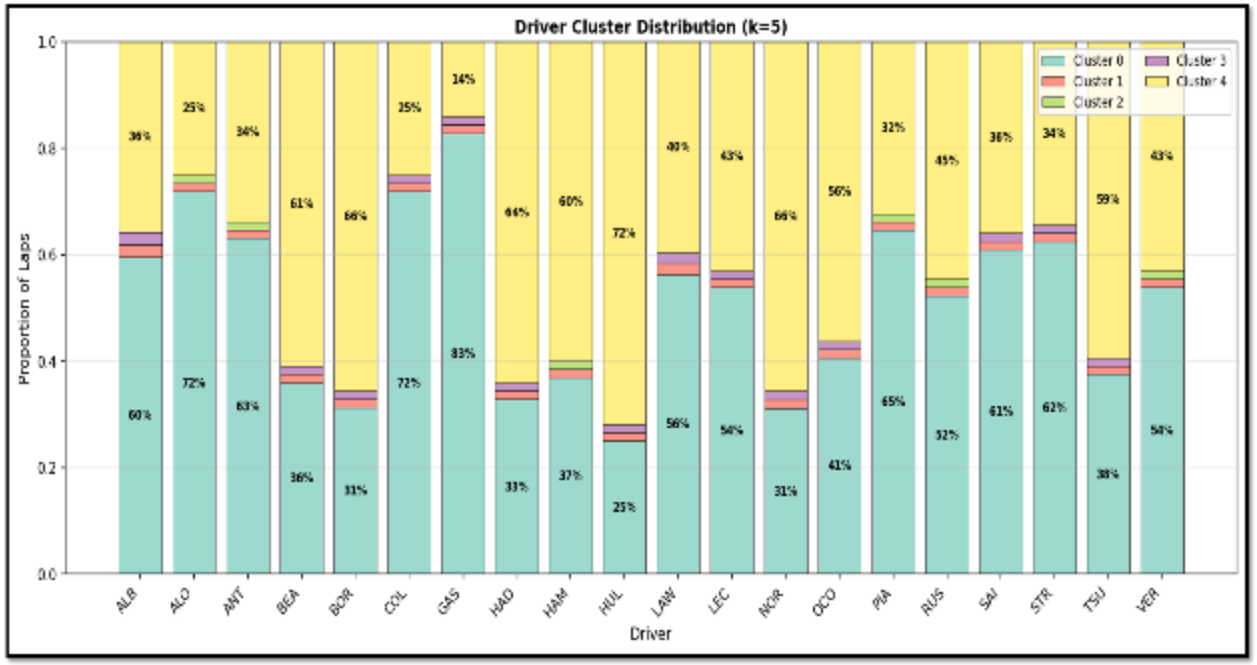
Data Normalization

- Scaled all telemetry metrics to a [0, 1] range
- Applied a 5-lap rolling window for local min-max normalization
- Removed mechanical bias from faster / slower cars

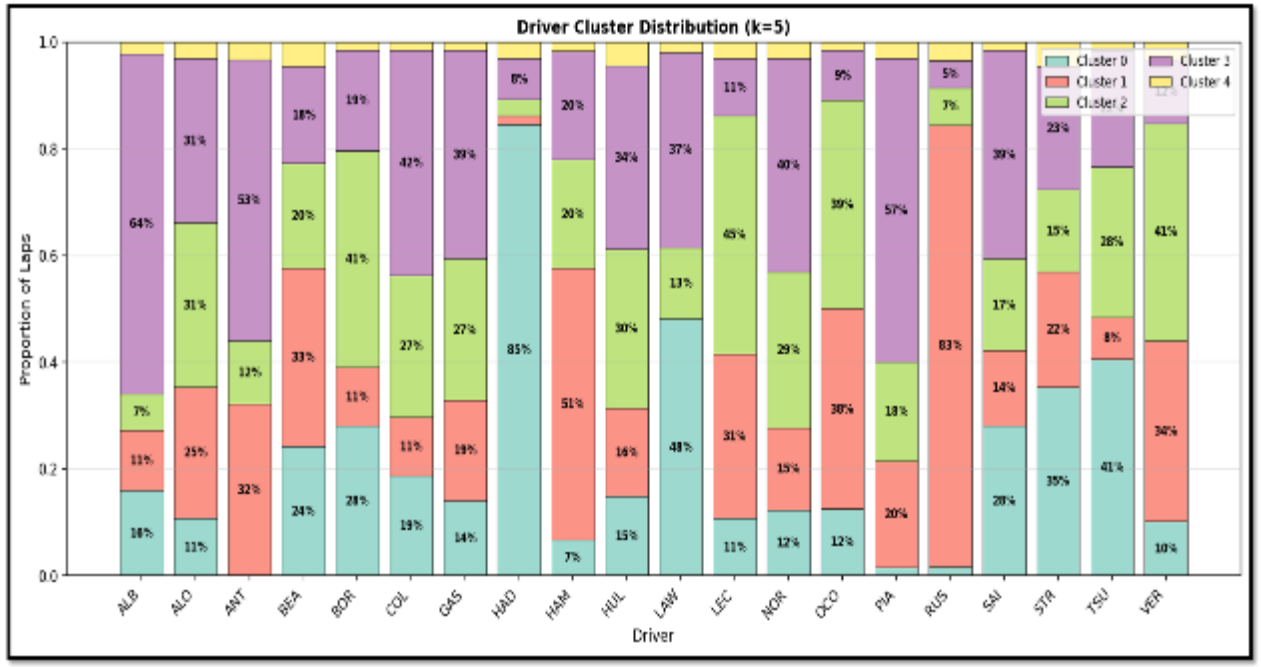
Outlier Detection

- Dropped laps with low average speed
- Used a 1.5 x IQR for a lower bound
- Detected pit laps, caution laps, laps with strange traffic

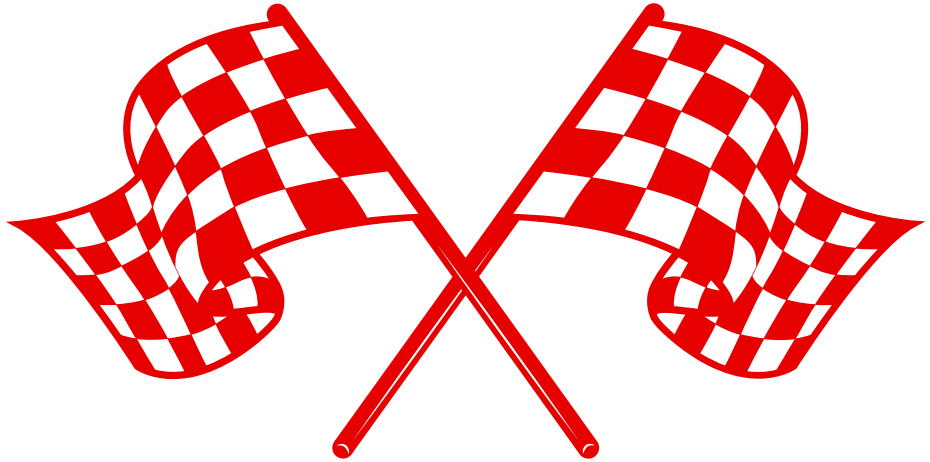
Dimensionality Reduction (PCA)



Without PCA



With PCA



Clustering

Clustering Algorithm Selection

These options are the most common unsupervised machine learning algorithms

K-Means

SELECTED

✓ **STRENGTHS**

- Fast & scalable to 18M+ records
- Works well after PCA reduction
- Easy to tune cluster count (k)

✗ **LIMITATIONS**

- Requires pre-specifying k
- Assumes convex clusters

Silhouette: 0.0653 | Davies-Bouldin: 2.61

Hierarchical

TESTED

✓ **STRENGTHS**

- No need to pre-specify k
- Dendrogram reveals structure

✗ **LIMITATIONS**

- $O(n^2)$ memory — too slow for 18M+ records
- Cannot scale to full 2025 season
- Works best on smaller data sets

Ruled out: memory constraint

Gaussian

TESTED

✓ **STRENGTHS**

- Soft/probabilistic assignments
- Handles elliptical clusters

✗ **LIMITATIONS**

- Assumes Gaussian distributions
- Unstable on high-dim PCA space
- Higher compute cost

Ruled out: distribution assumption mismatch

DBSCAN

TESTED

✓ **STRENGTHS**

- No cluster count needed
- Auto-detects noise/outliers

✗ **LIMITATIONS**

- Struggles with varying density
- ε tuning sensitive — poor cluster consistency across tracks

Ruled out: inconsistent clusters per race

Four algorithms evaluated — K-Means selected for performance and interpretability

K-Means Clustering

Cluster Evaluation Methods:

- Elbow Method – identifies diminishing returns in cluster separation
- Silhouette Score – measures cluster cohesion vs separation
- Davies-Bouldin Index – evaluates cluster similarity (lower is better)

Our Findings after many experimental models

- K-Means at 6 clusters produced the most well-balanced clusters (~1-2 outlier groups per race)
- Clusters can vary wildly depending on the track
- Metrics can be misleading

Cluster Representation Examples

Cluster 1
High Aggression
High RPM, throttle, speed — peak-performance laps

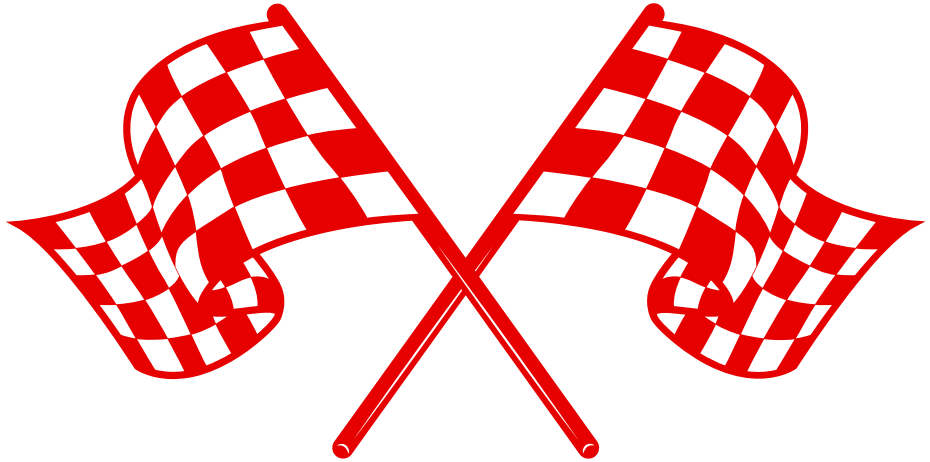
Cluster 2
Controlled Push
Elevated throttle, measured braking — raceable pace

Cluster 3
Balanced
Avg. across all metrics — representative race lap

Cluster 4
Cautious
Lower throttle/speed — caution laps or degradation

Cluster 5
Conservation
Low RPM, heavy coasting — tire/fuel management

We leave it up to the user to explore the data and label the groups



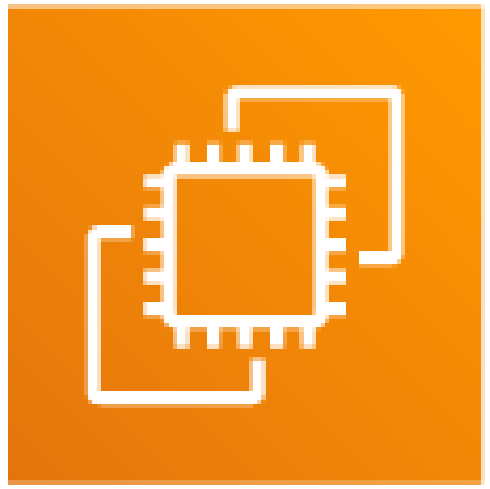
Front End

Deployment Stack

Accessed through a ***porkbun*** domain, served through ***NGINX***, hosted on an ***AWS EC2*** Instance.



porkbun



NGINX

Amazon EC2

Use Cases

Track Selection

Dropdown lists all **24** circuits from the 2025 season.



Driver Selection

Click on a driver to view their metrics and make comparisons.

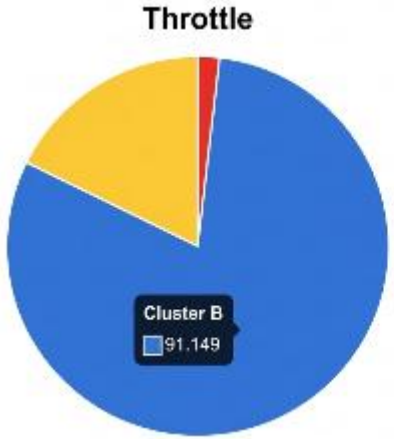


Statistics Exploration

One pie chart per metric.

Throttle, RPM, Gear, Brake, Acceleration, etc.

Radar Charts displaying **multivariate** clustering results.





Website

F1 Telemetry Insights Visual Dashboard

Driver Selection

Red Bull Racing	Ferrari	Mercedes	McLaren	Aston Martin	Alpine	Williams	Haas F1 Team	Racing Bulls	Kick Sauber
Verstappen	Leclerc	Russell	Norris	Alonso	Gasly	Sainz	Ocon	Tsunoda	Bortoleto
Lawson	Hamilton	Antonelli	Piastri	Stroll	Doohan	Albon	Bearman	Hadjar	Hülkenberg

Map Visualization: Select a track

Select a track to view telemetry map

Record

Updated Use Cases

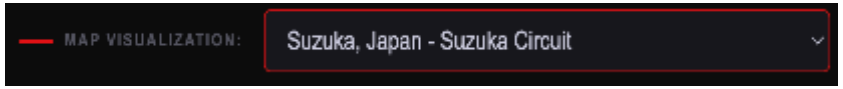
Driver Selection

Click on a driver to view their metrics and make comparisons.



Track Selection

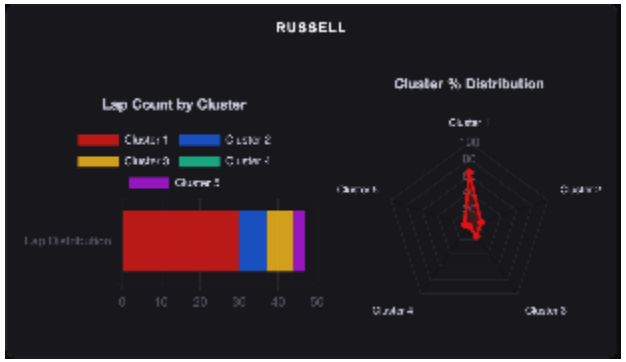
Dropdown menu updates map visualization and statistics block



Statistics Exploration

Bar chart shows lap count by cluster

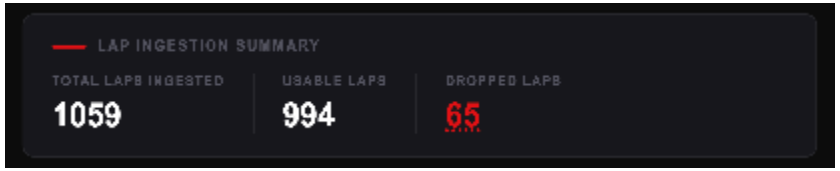
Radar Chart displays clustering distribution.

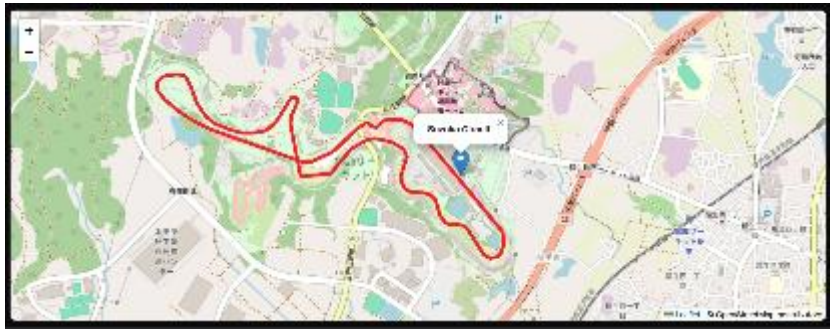




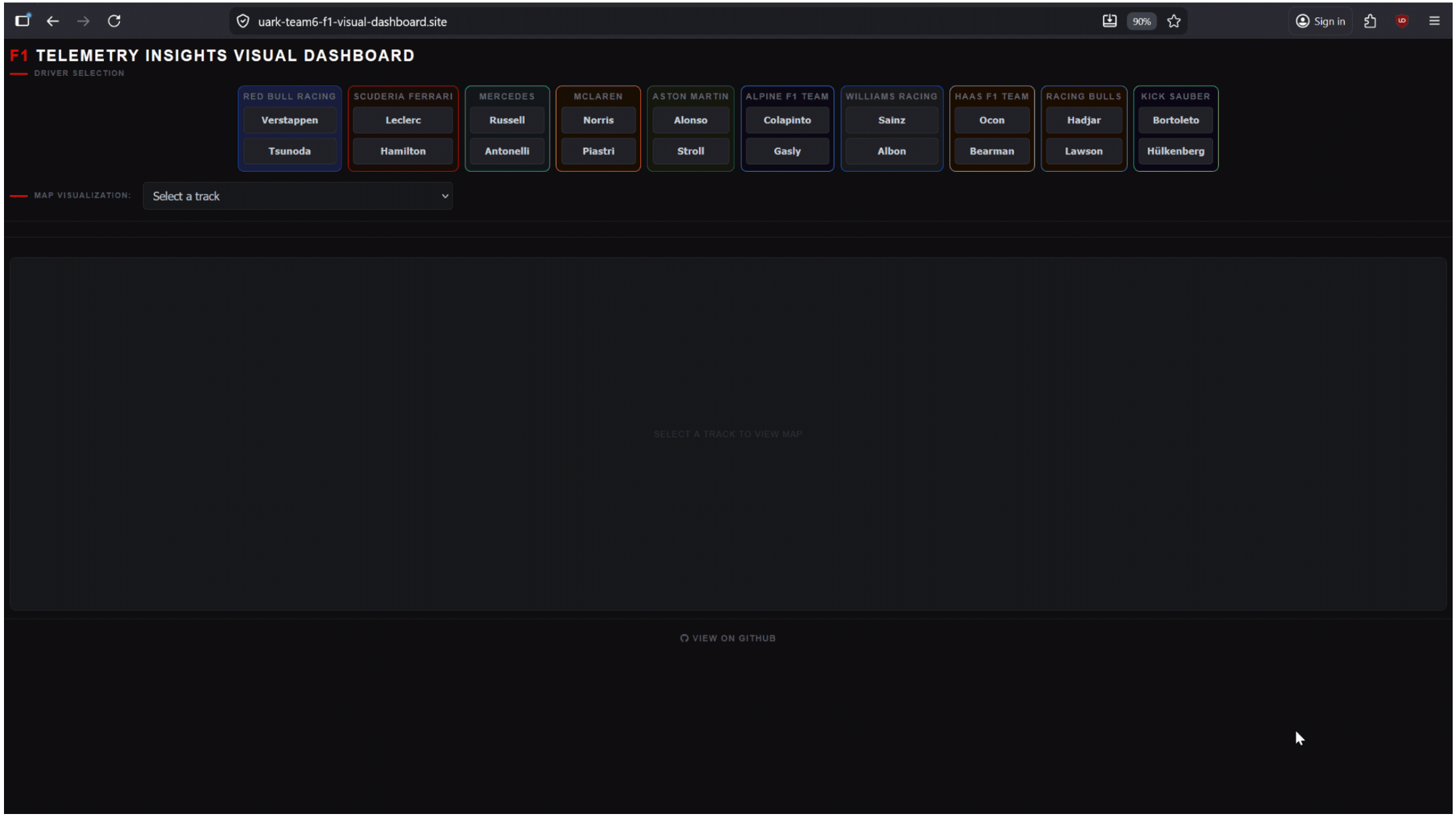
Map Visualization

Includes lap ingestion summary for further clustering analysis

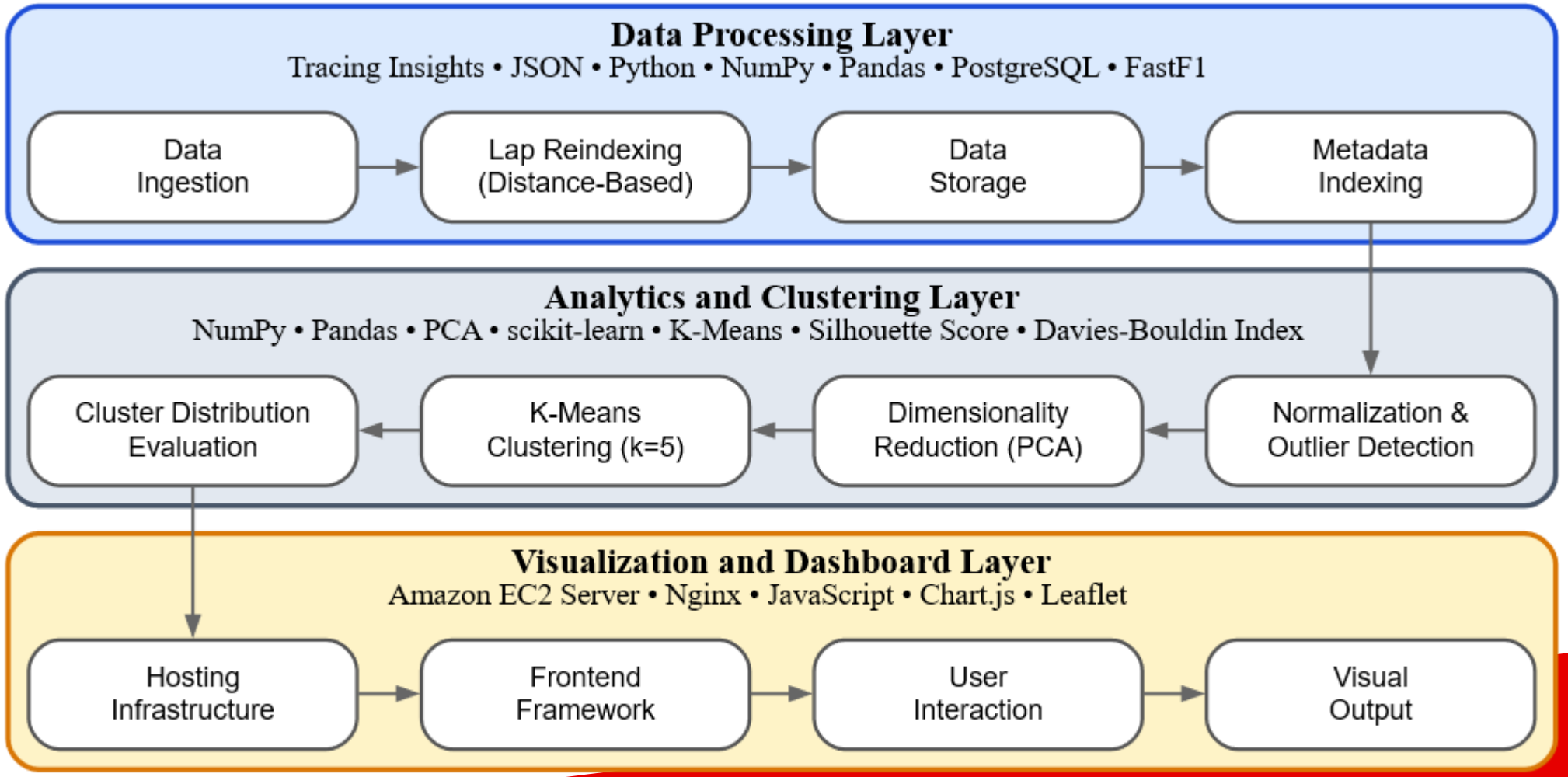




Website V2


uark-team6-f1-visual-dashboard.site
90%
Sign in
F1 TELEMETRY INSIGHTS VISUAL DASHBOARD
DRIVER SELECTION
RED BULL RACING
Verstappen
Tsunoda
SCUDERIA FERRARI
Leclerc
Hamilton
MERCEDES
Russell
Antonelli
MCLAREN
Norris
Piastri
ASTON MARTIN
Alonso
Stroll
ALPINE F1 TEAM
Colapinto
Gasly
WILLIAMS RACING
Sainz
Albon
HAAS F1 TEAM
Ocon
Bearman
RACING BULLS
Hadjar
Lawson
KICK SAUBER
Bortoleto
Hülkenberg
MAP VISUALIZATION:
Select a track
SELECT A TRACK TO VIEW MAP
VIEW ON GITHUB

Architecture



Formula 1

Telemetry Insights Visual Dashboard

MOTORSPORTS



All the way back to the starting line...